

# Navin Sridhar

---

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| CONTACT INFO                   | Department of Physics and KIPAC,<br>452 Lomita Mall, Stanford, CA 94305<br>E-mail: <a href="mailto:nsridhar@stanford.edu">nsridhar@stanford.edu</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| EMPLOYMENT                     | <b>Stanford University, CA</b> <ul style="list-style-type: none"><li>• Simons Collaboration (SCEECs) postdoctoral fellowship/KIPAC (2024–2028)</li><li>• Supervisor: Dr. Roger Blandford</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| EDUCATION                      | <b>Columbia University, NY</b> <ul style="list-style-type: none"><li>• Ph.D. in Astrophysics (May 2024)</li><li>• M.A. (2020), M.Phil (2021) in Astrophysics<br/>Advisors: Dr. Lorenzo Sironi, Dr. Brian D. Metzger<br/>Thesis: <a href="#">Electromagnetic and Multimessenger Signals From Magnetized Outflows of Compact Object Binaries</a></li></ul> <b>Indian Institute of Science Education &amp; Research, Bhopal</b> <ul style="list-style-type: none"><li>• BS-MS in Physics (CPI: 9.05/10) (May 2018)<br/>Thesis: Accretion disk evolution around black holes, and thermonuclear bursts in neutron stars. (Caltech and TIFR)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| AWARDS AND FELLOWSHIPS         | <ol style="list-style-type: none"><li>1. PI of NASA NICER grant (US\$45,000) (2026–2027)</li><li>2. Neil Gehrels Prize Fellowship: JSI, University of Maryland &amp; NASA (2024)</li><li>3. Brinson Fellowship: Northwestern University/CIERA &amp; University of Chicago (2024)</li><li>4. NASA <i>FINESST</i> award (US\$100,000) (2022–2024)</li><li>5. PI of NASA NICER grant (US\$37,500) (2022–2024)</li><li>6. Simons Foundation Aspen center for physics grant (US\$1,350) (July 2023)</li><li>7. International Astronomical Union (S378) travel grant (US\$1,600) (March 2023)</li><li>8. Columbia Arts and Sciences Graduate Council travel grant (US\$500) (2022)</li><li>9. Columbia ASGC Student/Diversity Initiative Grant (US\$2,000) (2022–2023)</li><li>10. AAS National Osterbrock Leadership fellow (US\$6,400) (2021–2023)</li><li>11. The Yeh Family endowment fellowship, Columbia University (2019–2020)</li><li>12. Dean’s fellowship for graduate study at Columbia University (2018–2020)</li><li>13. Caltech Summer Undergraduate Research Fellowship (US\$6,600) (July–Oct 2017)</li><li>14. Mitacs Globalink Research Internship Award (~CA\$6,600) (May–July 2017)</li><li>15. <i>INSPIRE</i> fellowship by DST, Govt. of India (INR 400,000) (2013–2018)</li></ol>                                                                                                               |  |
| RESEARCH EXPERIENCE/<br>VISITS | <ul style="list-style-type: none"><li>• Affiliated scientist, <a href="#">SCEECs</a> (2023–2028)</li><li>• <b>Simons</b> Collaboration on Extreme Electrodynamics of Compact Sources</li><li>• Visiting Researcher, <b>KITP</b>, UC Santa Barbara (Sept 2024)</li><li>• Visiting Student Researcher, <b>Caltech</b>, Pasadena (2023–2024)<br/>Host: Dr. Sterl Phinney</li><li>• Stellar Interactions and the Transients they Cause, <b>Aspen</b> (July 2023)<br/>Host: Aspen Center for Physics, Colorado</li><li>• Visiting researcher, Simons Foundation – plasma astrophysics program (2019–2023)<br/>Center for Computational Astrophysics, <b>Flatiron Institute</b>, NY<br/>Advisor(s): Dr. Sasha Philippov, Dr. Lorenzo Sironi</li><li>• Summer Undergraduate Research Fellow (SURF), <b>Caltech</b>, Pasadena (July–Oct 2017)<br/>Advisor(s): Dr. Javier García, Prof. Fiona Harrison</li><li>• Mitacs Globalink Research Intern, <b>University of Alberta</b>, Edmonton (May–July 2017)<br/>Advisor: Dr. Rodrigo Fernández</li><li>• Visiting Summer Research Program, <b>TIFR</b>, Mumbai (May–July 2016)<br/>Advisor: Prof. Sudip Bhattacharyya</li><li>• Indian Academy of Sciences summer research fellow, <b>ISRO</b>, Bangalore (June–Aug 2015)<br/>Advisor: Dr. Anuj Nandi</li><li>• Visiting Student Program, <b>IIAp</b>, Bangalore (May–Dec 2015)<br/>Advisor: Prof. Aruna Goswami</li></ul> |  |

1. **Sridhar, N.**, Sobacchi, E., Blandford, R., Sironi, L. 2026. “*Interaction of Strong X-mode and Fast Waves with Pair Plasmas*” to be submitted.
2. **Sridhar, N.**, Sobacchi, E., Sironi, L., et al. 2026. “*Interaction of Strong Electromagnetic Waves with Unmagnetized Pair Plasmas*” Physical Review Letters 137, 045102.
3. **Sridhar, N.**, Ripperda, B., Sironi, L., et al. 2025. “*Bulk Motions in the Black Hole Jet Sheath as a Candidate for the Comptonizing Corona*” ApJ 979, 199
4. **Sridhar, N.**, Metzger, B. D., & Fang, K. 2024. “*High-Energy Neutrinos from Gamma-Ray-Faint Accretion-Powered Hypernebulæ.*” ApJ 960, 74
5. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2023. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ II: Electron-Ion Plasma.*” MNRAS 518, 1301-1315
6. **Sridhar, N.**, & Metzger, B. D. 2022. “*Radio Nebulæ from Hyper-Accreting X-ray Binaries as Common Envelope Precursors and Persistent Counterparts of FRBs*” ApJ 937, 5
7. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2021. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ I: Magnetically Dominated Pair Plasma*” MNRAS 507, 5625-5640
8. **Sridhar, N.**, Metzger, B. D., Beniamini, Paz., et al. 2021. “*Periodic Fast Radio Bursts from Luminous X-ray Binaries.*” ApJ 917, 13
9. **Sridhar, N.**, Zrake, J., Metzger, B. D., et al. 2021. “*Shock-powered radio precursors of neutron star mergers from accelerating relativistic binary winds.*” MNRAS 501, 3184-3202
10. **Sridhar, N.**, García, J. A., Steiner, J. F., et al. 2020. “*Evolution of the Accretion Disk—Corona during the Bright Hard-to-soft State Transition: A Reflection Spectroscopic Study with GX 339-4.*” ApJ 890, 53
11. **Sridhar, N.**, Bhattacharyya, S., Chandra, S., & Antia, H. M. 2019. “*Broad-band reflection spectroscopy of MAXI J1535-571 using AstroSat: estimation of black hole mass and spin.*” MNRAS 487, 4221-4229

≤ 3<sup>rd</sup> author, with significant contributions (11): [<sup>†</sup> = supervised student]

12. Fan, N.<sup>†</sup>, & **Sridhar, N.** 2026. “*Comptonization by Shear-driven Bulk Motions in Black Hole Coronæ*” to be submitted to ApJ
13. Uno, K.<sup>†</sup>, Metzger, B. D., **Sridhar, N.** et al. 2026. “*The Population of Radio Hypernebulæ from Mass-Transferring Massive Binaries*” to be submitted to ApJ
14. Gupta, S.<sup>†</sup>, **Sridhar, N.**, Sironi, L. 2023. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ III: Dependence on the Guide Field in Pair Plasma.*” MNRAS 527, 6065-6075
15. Gupta, S.<sup>†</sup>, **Sridhar, N.**, Sironi, L. 2025. “*The Role of Electric Dominance for Particle Injection in Relativistic Reconnection*” MNRAS 538, 49-59
16. Bhandari, S., Marcote, B., **Sridhar, N.**, et al. 2023. “*Constraints on the persistent radio source associated with FRB 20190520B using the European VLBI Network*”, ApJ 958, 2
17. Eftekhari, T., Fong, W-F., Gordon, A. C., **Sridhar, N.**, et al. 2023. “*An X-ray census of Fast Radio Burst host galaxies: constraints on AGN and X-ray counterparts*”, ApJ 958, 1
18. Metzger, B. D., **Sridhar, N.**, Margalit, B., et al. 2022. “*A Toy Model for the Time-Frequency Structure of Fast Radio Bursts: Implications for the CHIME/FRB Burst Dichotomy.*” ApJ 925, 135
19. Margalit, B., Beniamini, P., **Sridhar, N.**, Metzger, B. D., 2020. “*Implications of a Fast Radio Burst from a Galactic Magnetar.*” ApJL 899, L27
20. García, J. A., Tomsick, J. A., **Sridhar, N.**, et al. 2019. “*The 2017 Failed Outburst of GX 339-4: Relativistic X-Ray Reflection near the Black Hole Revealed by NuSTAR and Swift Spectroscopy.*” ApJ 885, 48
21. Bhattacharyya, S., Yadav, J. S., **Sridhar, N.**, et al. 2018. “*Effects of Thermonuclear X-Ray Bursts on Non-burst Emissions in the Soft State of 4U 1728-34.*” ApJ 860, 88
22. Karinkuzhi, D., Goswami, A., **Sridhar, N.**, et al. 2018. “*Chemical analysis of three barium stars: HD 51959, HD 88035, and HD 121447.*” MNRAS 476, 3086-3096

> 3<sup>rd</sup> author (19):

23. Bhattacharyya, S., Dinsmore, J., ..., **Sridhar, N.**, et al. 2026. “*High-quality optical pulse profile and simultaneous X-ray pulsations from an accreting millisecond X-ray pulsar*” arXiv:2700.00000; Submitted to ApJ.

24. Adegoke, O., Bianchi, M., ..., **Sridhar, N.**, et al. 2026. “*Following the Flow: Accretion and X-Ray Reflection in the 2021 Outburst of GX 339-4*” arXiv:2700.00000; Submitted to ApJ.
25. Shuvo, P., Meyer, E., ..., **Sridhar, N.**, et al. 2026. “*First Detection of Radio Polarization During Jet Formation in the Changing-Look AGN 1ES 1927+654*” arXiv:2609.09539; Submitted to ApJ.
26. Uttarkar, P., Shannon, R., ..., **Sridhar, N.**, et al. 2026. “*A fast radio burst cyclone in technicolour: evidence of plasma lensing*” arXiv:2602.16409; Submitted to Nature Astronomy
27. Moroianu, A., Bhandari, S., ..., **Sridhar, N.**, et al. 2026. “*A Milliarcsecond Localization Associates FRB 20190417A with a Compact Persistent Radio Source and an Extreme Magnetoionic Environment*” ApJL 996, L16
28. Snelders, M. P., Hessels, J. W. T., ..., **Sridhar, N.**, et al. 2025. “*Revisiting FRB 20121102A: milliarcsecond localisation and a decreasing dispersion measure*” arXiv:2510.11352
29. Blanchard, P., Berger, E., ..., **Sridhar, N.**, et al. 2025. “*James Webb Space Telescope Observations of the Nearby and Precisely Localized FRB20250316A: A Potential Near-IR Counterpart and Implications for the Progenitors of Fast Radio Bursts*” ApJL 989, L49
30. Ibik, A., Drout, M., ..., **Sridhar, N.**, et al. 2024. “*A search for persistent radio sources toward repeating fast radio bursts discovered by CHIME/FRB*” ApJ 976, 199
31. Dong, Y., Eftekhari, T., ..., **Sridhar, N.**, et al. 2024. “*A Radio Study of Persistent Radio Sources in Nearby Dwarf Galaxies: Implications for Fast Radio Bursts*”, ApJ 973, 133
32. Sarin, N., Clarke, T. A., ..., **Sridhar, N.** 2024. “*The origin of the coherent radio flash potentially associated with GRB 201006A*”, ApJL 973, L20
33. AXIS TDAMM SWG., ..., incl. **Sridhar, N.**, et al. 2023. “*Prospects for Time-Domain and Multi-Messenger Science with AXIS*”, Universe, 10, 316
34. Connors, R. M. T., Tomsick, J. T., ..., **Sridhar, N.**, et al. 2023. “*The High Energy X-ray Probe (HEX-P): Probing Accretion onto Stellar Mass Black Holes*”, Front. Astron. Space Sci. 10:1292682
35. Kammoun, E., Lohfink, A. M., ..., **Sridhar, N.**, et al. 2023. “*The High Energy X-ray Probe (HEX-P): Probing the physics of the X-ray corona in active galactic nuclei*”, Front. Astron. Space Sci. 10:1308056
36. Dong, Y., Eftekhari, T., ..., **Sridhar, N.**, et al. 2023. “*Mapping Obscured Star Formation in the Host Galaxy of FRB 20201124A*”, ApJ 961, 44
37. Bhandari, S., Gordon, A. C., ..., **Sridhar, N.**, et al. 2023. “*A non-repeating fast radio burst in a dwarf host galaxy*”, ApJ 948, 67
38. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2021. “*Reflection Modeling of Black Hole Binary 4U 1630–47: Disk Density and Returning Radiation*”, ApJ 909, 146
39. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2020. “*Evidence for Returning Disk Radiation in the Black Hole X-Ray Binary XTE J1550–564.*” ApJ 892, 47
40. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2019. “*Conflicting Disk Inclination Estimates for the Black Hole X-Ray Binary XTE J1550–564.*” ApJ 882, 179
41. Verdhnan Chauhan, J., Yadav, J. S., ..., **Sridhar, N.**, et al. 2017. “*AstroSat/LAXPC Detection of Millisecond Phenomena in 4U 1728-34.*” ApJ 841, 41

Books and white papers (2):

42. Koss, M., ., incl. **Sridhar, N.**, et al. 2025. “*The Advanced X-ray Imaging Satellite Community Science Book*”, arXiv:2511.00253
43. Safi-Harb, S., Burdge, K., ..., incl. **Sridhar, N.**, et al. 2023. “*From Stellar Death to Cosmic Revelations: Zooming in on Compact Objects, Relativistic Outflows and Supernova Remnants with AXIS*”, arXiv:2311.07673

Astronomical Telegrams (2):

44. Blanchard, P., Berger, E., ..., **Sridhar, N.**, et al. 2015. “*Deep JWST/NIRCam Imaging of FRB 20250316A: Detection of Potential IR Counterparts.*” ATel #17195
45. García, J. A., Harrison, F., ..., **Sridhar, N.**, et al. 2017. “*NuSTAR Observation of GX 339–4 in the early stages of its 2017 outburst.*” ATel #10825

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBSERVING  | X-ray: 15 Ms; Radio: 142 ks; Optical: 65 ks; IR: 4.5 hr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| EXPERIENCE | Total telescope grant: \$624,935                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|            | <u>As Principal Investigator:</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|            | <ol style="list-style-type: none"> <li>1. NICER archival data of transitional ms pulsar (Cycle AO8; 1.2 Ms of data; US\$45,000).</li> <li>2. Joint NICER and NuSTAR observations studying reflection features from X-ray binaries (Cycle AO3; 60 ks; US\$37,500).</li> <li>3. uGMRT multi-band radio observations of Giant Radio Pulses (Cycle 34; 36 ks).</li> <li>4. <i>AstroSat</i> observations of <math>\sim 10</math> sources: accretion powered millisecond pulsars, neutron star type-I bursts, black hole X-ray binaries (Cycles A04-A08; cumulatively &gt;300 ks).</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|            | <u>As Co-Investigator:</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|            | <ol style="list-style-type: none"> <li>1. JWST search for a transient NIR counterpart to FRB 20250316A (PI: Peter Blanchard; \$110,500 ; cycle 5, 2026).</li> <li>2. JWST observation of FRB 20250316A (PI: Peter Blanchard; cycle 4, 2025).</li> <li>3. Joint NuSTAR and NICER observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle 10; 500 ks; PI: J. Garcia).</li> <li>4. GBT observations to probe the dynamic environment of FRB 20180301A (Cycle 2024B; 108 ks; PI: P. Kumar).</li> <li>5. Chandra observations to probe the engines of fast radio bursts (Cycle 25; 106 ks; PI: T. Eftekhari).</li> <li>6. GBT observations to probe the local environment of FRB 20180301A (Cycle 2023B; 72 ks; PI: P. Kumar).</li> <li>7. VLA observations to constrain FRB ‘persistent radio sources’ (Cycle 2023B; 10 ks; PI: Y. Dong).</li> <li>8. NICER observations for amassing a comprehensive spectral timing archive of outbursting black holes (Cycle AO5; 200 ks; \$43,000; PI: J. Steiner).</li> <li>9. Joint NICER and NuSTAR observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle AO4; 110 ks; \$44,000 ; PI: J. Garcia).</li> <li>10. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO4; 88 ks; \$44,000; PI: R. Connors).</li> <li>11. NICER observations for black hole spin measurements using continuum fitting (Cycle AO4; 40 ks; \$37,500; PI: J. Steiner).</li> <li>12. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO4; 30 ks; PI: G. Mastroserio).</li> <li>13. Joint XMM-Newton and NuSTAR observations for studying truncated accretion disks in black holes (AO 21; 120 ks; \$50,656; PI: J. García).</li> <li>14. Chandra observation of nearby Fast Radio Bursts (Cycle 23; 55 ks; \$52,519; PI: T. Eftekhari).</li> <li>15. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO3; 88 ks; \$37,500; PI: R. Connors).</li> <li>16. NICER observations for black hole spin measurements using continuum fitting (Cycle AO3; 40 ks; \$44,000; PI: J. Steiner).</li> <li>17. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO3; 30 ks; \$37,500; PI: G. Mastroserio).</li> <li>18. NICER observations monitoring the evolving thermal disk emission of LMC X-3 (Cycle AO2; 60 ks; \$20,229; PI: J. Steiner).</li> <li>19. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO2; 90 ks; \$21,031; PI: R. Connors).</li> </ol> |
|            | <u>Others:</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|            | <ol style="list-style-type: none"> <li>1. Mt. Palomar 200” Hale telescope: DBSP observations of 14 variable AGN (2 nights)</li> <li>2. NRAO Green Bank Telescope: radio pulsar observations (24 ks)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

DELIVERED

TALKS/

PRESENTATIONS

(\*32 INVITED)

86. \*Talk at the PCTS workshop on Black Hole corona 2027 **Princeton University**, NJ (March 2027)
85. \*Talk at the Black Hole corona workshop 2026 **Princeton University**, NJ (Oct 2026)
84. Talk at the Kavli Astrophysics Symposium 2026 **Cambridge**, UK (Sept 2026)
83. Talk and \*discussion session lead at FRB 2026 conference, **Guiyang**, China (July 2026)
82. \*Seminar at Department of Astronomy, **Tsinghua University**, Beijing (July 2026)
81. Talk at the BlackHolic 2026 conference, **Oxford University**, UK (March 2026)
80. \*Astrophysics seminar, **Washington University**, St. Louis, MO (Oct 2025)
79. 22nd AAS-High Energy Astrophysics Division meeting, St. Louis, MO (Oct 2025)
78. \*Black-board talk, Frontiers of Relativistic Plasma Physics, **KITP**, USA (Sep 2025)
77. \*Overview of FRB emission mechanisms, FRB 2025 conference, **McGill**, Canada (July 2025)
76. Seminar, **Inter-University Center for Astronomy & Astrophysics**, Pune (June 2025)
75. \*Astrophysics colloquium, **Tata Institute of Fundamental Research**, Mumbai (May 2025)
74. Astrophysics and Relativity seminar, **International Center for Theoretical Sciences**, Bangalore (May 2025)
73. Relativistic Magnetospheres workshop, **Ecole De Physique Des Houches**, France (April 2025)
72. Magnetic fields around Compact Objects workshop, **Perimeter Institute**, Canada (March 2025)
71. Fast Radio Burst Frontiers workshop, **Carnegie Mellon University**, USA (March 2025)
70. \*Talk at the XOC group meeting **Stanford**, CA (Sept 2024)
69. Talk at the High-energy plasma phenomena in astrophysics workshop **MIAPbP**, Munich, Germany (Sept 2024)
68. Graduating student colloquium, **Columbia University**, NY (April 2024)
67. \***AXIS** seminar series (April 2024)
66. 21st AAS-High Energy Astrophysics Division meeting, Texas (April 2024)
65. \*Seminar, Simons Collaboration on Extreme Electrodynamics of Compact Object Sources (**SCEECS**) (Feb 2024)
64. \*Astroplasmas seminar, **Princeton University**, NJ (Dec 2023)
63. \*CIERA Theory seminar, **Northwestern University**, IL (Nov 2023)
62. JSI workshop on ‘Winds throughout the Universe’, **Annapolis**, MD (Oct 2023)
61. \*Theoretical astrophysics group board talk, **Carnegie Observatories**, CA (Sept 2023)
60. \*Seminar, **Syracuse University**, NY (Sept 2023)
59. \*Review talk at Astrophysics of Fast Radio Bursts II workshop (+session chair), **CCA, Flatiron Institute**, NY (Sept 2023)
58. \*Colloquium, **National Center for Radio Astrophysics**, Pune, India (June 2023)
57. \*Astrophysics seminar, **Tata Institute of Fundamental Research**, Mumbai (May 2023)
56. \*‘Transient Tuesday’ talk, DARK, Niels Bohr Institute **Copenhagen**, Denmark (May 2023)
55. \*Discussion chair, Fast Radio Burst Follow-Up workshop, **University of Toronto** (April 2023)
54. \*High energy astrophysics seminar, **Hebrew University of Jerusalem**, Israel (March 2023)
53. \*Seminar, Astrophysics Research Center of **The Open University**, Israel (March 2023)
52. \*High energy astrophysics meeting, Black Hole Initiative, **Harvard**, MA (Feb 2023)
51. Cosmic transients lab seminar, Center for Astrophysics, **Harvard**, MA (Feb 2023)
50. High Energy Astrophysics seminar, Center for Astrophysics, **Harvard**, MA (Feb 2023)
49. FRB Taiwan 2023, National Chung Hsing University, Taichung **Taiwan** (Feb 2023)
48. \*Workshop on ‘Overcoming disconnects in the understanding of accreting black holes’, **Lorentz Center, Leiden**, The Netherlands (Jan 2023)
47. HEX-P black hole binaries, **science working group** presentation (Dec 2022)
46. \*CCAPP seminar, **Ohio State University**, OH (Nov 2022)
45. 64th APS Division of Plasma Physics annual meeting, **Spokane**, WA (Oct 2022)
44. Review talk on Observations of FRBs, **CCA, Flatiron Institute**, NY (Oct 2022)
43. \***AXIS** ‘compact objects and supernova remnants’ **science working group** (Oct 2022)
42. \*Astrophysics seminar, **Cornell University**, NY (Sept 2022)
41. \*AGN seminar at **NASA** Goddard Space Flight Center, MD (Sept 2022)
40. **NASA** Time Domain and Multi-Messenger (TDAMM) Initiative Workshop, **Annapolis**, MD (Aug 2022)
39. FRB 2022 meeting, IAUGA, **Busan**, South Korea (Aug 2022)
38. Summer BLAST workshop, Black Hole Initiative, **Harvard**, MA (July 2021)
37. **AXIS** ‘Time-Domain and Multi-Messenger Astronomy’ **science working group** (July 2022)
36. Special pizza lunch seminar, **Caltech**, CA (June 2022)
35. Seminar at **Carnegie Observatories**, CA (June 2022)
34. 240th meeting of the American Astronomical Society, **Pasadena**, CA (June 2022)

33. Workshop on Relativistic Plasma Astrophysics, **Purdue University**, IN (May 2022)
32. Compact objects group meeting at **CCA, Flatiron Institute**, NY (April 2022)
31. \*Kapteyn Astronomical Institute, **Groningen**, The Netherlands (April 2022)
30. 19th AAS-High Energy Astrophysics Division meeting, **Pittsburgh**, PA (March 2022)
29. Brown bag lunch seminar, **Massachusetts Institute of Technology**, MA (Feb 2022)
28. Winter BLAST workshop, Black Hole Initiative, **Harvard**, MA (Dec 2021)
27. Gothamfest, **CCA, Flatiron Institute**, NY (Dec 2021)
26. Kavli-IPMU workshop on particle acceleration by reconnection, **Tokyo**, Japan (Nov 2021)
25. 9th Microquasar workshop, **Cagliari**, Italy (Sept 2021)
24. Astrofest, **Columbia University**, NY (Sept 2021)
23. High-energy plasma phenomena in astrophysics, **Munich**, Germany (July 2021)
22. FRB 2021 conference (online) (July-Aug 2021)
21. \*Princeton astro-coffee, **Princeton**, NJ (July 2021)
20. ‘Midwest Magnetic Fields’ Workshop, **Wisconsin-Madison**, (June 2021)
19. \*CHIME/FRB collaboration, **McGill**, Canada (May 2021)
18. \*Seminar at the **Institute of Astrophysics - FORTH**, Greece (April 2021)
17. Compact objects group meeting at **CCA, Flatiron Institute**, NY (March 2021)
16. **Kyoto** FRB meeting, **Yukawa Institute for Theoretical Physics** (Feb 2021)
15. Compact object group meeting, **TIFR**, Mumbai (Dec 2020)
14. 62nd Annual Meeting of the **APS Division of Plasma Physics** (Nov 2020)
13. Compact objects group meeting, **CCA, Flatiron Institute**, NY (June, Oct, Nov 2020)
12. Plasma astrophysics summer program, **CCA, Flatiron Institute**, NY (July 2019)
11. Theoretical High Energy Astrophysics seminar, **Columbia University** (June 2019)
10. Graduate Seminar, **Columbia University**, NY (Feb, April 2019)
9. Research Seminar, **Columbia University**, NY (Dec 2018)
8. Pizza lunch talk, **Columbia University**, NY (Oct 2018, 2022)
7. *NuSTAR* SOC group meeting, **Caltech**, CA (Sept 2017)
6. Group presentation at **NRAO Green Bank Observatory**, WV (Sept 2017)
5. 16th AAS-High Energy Astrophysics Division meeting, **Sun Valley**, ID (March 2022)
4. Visiting Student Program ceremony, **University of Alberta**, Canada (July 2017)
3. Lunch talk at **University of Alberta**, Canada (June 2017)
2. ‘Wide band X-ray spectro-temporal studies’, **TIFR**, Mumbai, India (Jan 2017)
1. **IISER Bhopal** Physics department symposium (Nov 2015)

MENTORING Total: 10 students; 4 papers.

At Stanford University:

- *Shanzay Sajid*; **High-school** student, Lahore Grammar School (Fall 2026-)
- *Ningyue Fan*; **Ph.D.** student, Stanford University (Fall 2025-)
- *Christopher Farias*; **undergraduate** student, San Diego State University (Summer 2025)

At Columbia University:

- *Sanya Gupta*; **undergraduate** research projects (2 papers) → Ph.D. student at Stanford University (2022–2025)
- *Kohki Uno*; **Ph.D.** student, Kyoto University (1 paper) → postdoc at Columbia University (Summer 2024)
- *Shifra Mandel*; **graduate** mentor-mentee program (2021–2022)
- *Arnab Kundu*; **masters** thesis student, IISER Kolkata (2021–2022)
- *Akash Vani*; Pune University → Masters at Heidelberg University → Ph.D. student at MPA, Garching (2019–2020)
- *Shuhan Zhang*; Women in Science at Columbia **undergraduate** mentorship program → Ph.D. student at UC San Diego (2021)
- *Bhairavi Chandrasekhar*; **undergraduate** summer internship (2020)

|                         |                                                                                                                                                                                                                                           |                                          |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| TEACHING                | At Columbia University:                                                                                                                                                                                                                   |                                          |
|                         | – <b>Head TA</b> of the Astronomy department                                                                                                                                                                                              | (2020–2021)                              |
|                         | – <b>Instructor</b> for undergraduate Intro to Astronomy lab courses                                                                                                                                                                      | (Fall 2019, Spring 2020)                 |
|                         | – <b>Teaching Assistant</b> for 2 undergraduate (non-major) courses                                                                                                                                                                       | (Fall 2018, Spring 2019)                 |
|                         | – <b>Guest lectured</b> for ASTR UN1403 (in lieu of Prof. David Helfand)                                                                                                                                                                  | (Feb 2020)                               |
|                         | – <b>Advanced track</b> in Columbia CTL’s <a href="#">Teaching Development Program (TDP)</a>                                                                                                                                              | (2018–2024)                              |
|                         | – Completed CTL’s <a href="#">Innovative Course Design Seminar</a>                                                                                                                                                                        | (2021)                                   |
|                         | Outside Columbia University:                                                                                                                                                                                                              |                                          |
|                         | – Selected as a <b>Lead instructor</b> at the Yale Young Global Scholars (YYGS) program                                                                                                                                                   |                                          |
|                         | <b>Yale University</b> , New Haven, CT                                                                                                                                                                                                    | (Summer 2020; postponed due to Covid-19) |
| PROFESSIONAL LEADERSHIP | – Trained to incorporate <b>inquiry-based learning techniques</b> in STEM education through the <a href="#">ISEE—Professional Development Program (PDP)</a> , <b>UC Santa Cruz</b>                                                        | (March–Sept 2019)                        |
|                         | – <a href="#">Designed and taught an inquiry activity</a> on statistics, <b>UC Los Angeles</b>                                                                                                                                            | (Sept 2019)                              |
|                         | • Scientific Organizing Committee of symposium on <i>The birth, life and death of black holes</i> , European Astronomical Society annual meeting, Leiden, The Netherlands                                                                 | (June 2021)                              |
|                         | At Stanford University:                                                                                                                                                                                                                   |                                          |
|                         | • Postbaccalaureate program admissions committee member                                                                                                                                                                                   | (2025)                                   |
|                         | • Ph.D. program admissions committee member                                                                                                                                                                                               | (2024–2025)                              |
|                         | • Started the weekly <a href="#">KIPAC Compact Objects Group meetings</a>                                                                                                                                                                 | (Oct 2024–)                              |
|                         | At Columbia University:                                                                                                                                                                                                                   |                                          |
|                         | • Organized an eight-part lecture series titled <i>‘Science Policy for Scientists’</i>                                                                                                                                                    | (Spring 2023)                            |
|                         | • Department representative, Arts and Sciences Graduate Council                                                                                                                                                                           | (2022–2023)                              |
| PROFESSIONAL SERVICE    | • Graduate representative, faculty hiring committee, Astronomy department                                                                                                                                                                 | (2021–2022)                              |
|                         | • Graduate representative, colloquium committee, Astronomy department                                                                                                                                                                     | (2022–2023)                              |
|                         | • Graduate admissions committee member, Astronomy department                                                                                                                                                                              | (2021–2022)                              |
|                         | • Co-convenor of <i>Astrofest</i>                                                                                                                                                                                                         | (Sept 2020)                              |
|                         | At IISER Bhopal:                                                                                                                                                                                                                          |                                          |
|                         | • Senate representative, Central Advisory Committee, Students Activity Council                                                                                                                                                            | (2016–2017)                              |
|                         | • Elected Science Council secretary                                                                                                                                                                                                       | (2014–2016)                              |
|                         | • Convenor of <i>Singularity</i> , the Institute annual science festival                                                                                                                                                                  | (April 2015)                             |
|                         | • Founder of the Institute <i>Astronomy Club</i>                                                                                                                                                                                          | (2014)                                   |
|                         | • Founder of the Institute <i>Quiz Club</i>                                                                                                                                                                                               | (2014)                                   |
|                         | • Invited to serve as a <b>referee</b> for:                                                                                                                                                                                               |                                          |
|                         | –Journal of Plasma Physics,                                                                                                                                                                                                               |                                          |
|                         | –Proceedings of the National Academy of Sciences of the USA,                                                                                                                                                                              |                                          |
|                         | –Nature,                                                                                                                                                                                                                                  |                                          |
|                         | –The Astrophysical Journal,                                                                                                                                                                                                               |                                          |
|                         | –Monthly Notices of Royal Astronomical Society,                                                                                                                                                                                           |                                          |
|                         | –Physical Review D, X,                                                                                                                                                                                                                    |                                          |
|                         | –Astronomy & Astrophysics.                                                                                                                                                                                                                |                                          |
|                         | • Invited to serve as a <b>panelist</b> for:                                                                                                                                                                                              |                                          |
|                         | –NASA’s Chandra, NICER observatories                                                                                                                                                                                                      |                                          |
|                         | –National Science Foundation,                                                                                                                                                                                                             |                                          |
|                         | – <i>AstroSat</i> ,                                                                                                                                                                                                                       |                                          |
|                         | –NCRA-uGMRT,                                                                                                                                                                                                                              |                                          |
|                         | –Observer at Hubble Space Telescope time allocation committee.                                                                                                                                                                            |                                          |
|                         | • <b>Science working group member</b> of:                                                                                                                                                                                                 |                                          |
|                         | –NewAthena                                                                                                                                                                                                                                | (2025–)                                  |
|                         | –Advanced X-ray Imaging Satellite (AXIS)                                                                                                                                                                                                  | (2022–)                                  |
|                         | –High-Energy X-ray Probe (HEX-P)                                                                                                                                                                                                          | (2022–)                                  |
|                         | • DEI committee member, Columbia Arts and Sciences Graduate Council                                                                                                                                                                       | (2022–2023)                              |
|                         | • <b>Judge</b> , Chambliss poster competition, American Astronomical Society meeting                                                                                                                                                      | (2022)                                   |
| SUMMER SCHOOLS          | • <b>Webmaster:</b> (1) <a href="#">Columbia Astronomy Department</a> , (2) <a href="#">Theoretical High-Energy Astrophysics group, Columbia</a> , (3) <a href="#">Simons Collaboration on Extreme Electrodynamics of Compact Sources</a> |                                          |
|                         | • Member of American Astronomical Society                                                                                                                                                                                                 | (2018–present)                           |
|                         | • Member of New York Academy of Sciences                                                                                                                                                                                                  | (2018–2023)                              |
|                         |                                                                                                                                                                                                                                           |                                          |



- **Multiscale modeling of astrophysical and space plasmas – 2019**  
Center for Computer Astrophysics (CCA), Flatiron Institute, New York, NY
- **Multiple approaches to plasma physics from laboratory to astrophysics – 2019**  
Ecole de Physique Des Houches, Les Houches, France
- **NRAO Green Bank Telescope observer training workshop – Fall 2017**  
Green Bank Observatory, West Virginia
- **Summer School on Gravitational Wave Astronomy – 2016**  
International Center for Theoretical Sciences (ICTS), TIFR, Bangalore
- **Summer school on relativity, black holes and neutron stars – 2015**  
Indian Institute of Astrophysics (IIAp), Bangalore

COMPUTING/  
SOFTWARE  
EXPERIENCE

- Languages and tools: Python, git, FORTRAN, IRAF, gnuplot, DS9, L<sup>A</sup>T<sub>E</sub>X, *f*tools (HEASoft): XSPEC, QDP, XRONOS
- Supercomputers: NASA Pleiades, DoE-NERSC Cori, NSERC WestGrid, Flatiron Institute HPC cluster, Columbia University Habanero, Terremoto, Ginsburg clusters
- Numerical tools: Particle-In-Cell, Magnetohydrodynamical simulations
- Telescope data handling: *AstroSat*, *Swift*, *RXTE*/PCA, *NuSTAR*, *NICER*, *XMM-Newton*, NCRA-GMRT, NRAO-GBT (Astrid, CLEO)

SCIENCE  
OUTREACH

1. Public outreach talk during KIPAC community day, Stanford University (April 2025)
2. Research career in astrophysics for undergraduates at IISER Bhopal (Jan 2025)
3. [KIPAC public outreach talk](#), Stanford University (Dec 2024)
4. Co-organized [astronomy outreach activities](#) at Columbia University for the South African academic achievers program (750 high-school students) (Oct 2022)
5. Talk on introduction/orientation to summer research, for undergraduate researchers at Columbia University (June 2022)
6. Popular talk on black hole X-ray binaries at IISER Bhopal Astronomy club (Oct 2021)
7. [Mentored the students of Democracy Prep High School](#), Harlem, NYC, on the foundations of computer coding (Fall 2021)
8. Research career in astrophysics for undergraduates at IISER Bhopal (April 2021)
9. Delivered a lecture on science with space-based X-ray telescopes, for high-school students at *Frontiers of Science*—the annual outreach program at TIFR, Mumbai. (Nov 2017)
10. Organized a tour of GMRT for TIFR–Visiting Summer Research students (June 2016)

News coverage and appearances:

- *Super sharp images with EVN reveal another Fast Radio Burst Coming from a Hypernebula.* [[JIVE](#); 29-Nov-2023][[ASTRON](#); 01-Dec-2023][[Universe Today](#); 05-Dec-2023]
- *Astronomers caught a potent radio burst blasting at us from a dwarf galaxy 3 billion light-years away* [[Popular Science](#); 09-June-2022]
- *AstroSat Catches Nuclear Reactions Spreading Across a Neutron Star* [[The Wire](#); 07-October-2021]
- *Long burst short: an unexpected supernova* [[Research Matters](#); 26-July-2021]
- *Fast Radio Bursts from globular clusters and what could power them* [[National Geographic](#); 27-May-2021]
- *Digging the grave: In search of the remains of merging black holes and neutron stars* [[Research Matters](#); 14-Sept-2021]
- *FRBs from Galactic magnetar* [[The Wire](#); 03-June-2020] [[AAS Nova](#); 09-Sept-2020]
- *Black hole bends light on itself* [[Gizmodo](#); 10-April-2020] [[Caltech News](#); 08-April-2020] [[SYFY](#); 09-July-2020]